

Convolutional Neural Network Based IOT Intrusion Detection System using Edge-IIoTset

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Abstract—The world of IoT security is rapidly evolving, especially with the rise of advanced technologies. The paper is a study of how Convolutional Neural Networks into Intrusion Detection Systems (IDS) for Edge Industrial Internet of Things (EdgeIIoT) datasets can greatly improve the security of IoT devices. By utilizing deep learning and CNNs in IoT security, we can develop advanced anomaly detection algorithms capable of identifying unusual patterns and behaviors that may indicate potential security threats. The primary benefit of utilizing CNNs for Intrusion Detection Systems (IDS) in EdgeIIoT security lies in their capacity to autonomously learn and extract features from unprocessed data. This capability proves especially valuable when managing the extensive streams of varied data produced by EdgeIIoT devices. This paper explores how CNN and its variations, Xception and VGG16 (Visual Geometry Group 16), help improve the security of an Edge-IIoTset. Generic CNN tops both Xception and VGG16 models with the highest accuracy of 98.98%. The vast field of IoT security is growing rapidly, especially because of advanced technologies. This case study delves into the application of CNNs in IDS for EdgeIIoT datasets, displaying a boost in security measures for IoT devices.

Keywords—Convolutional Neural Networks, VGG16, Xception, IoT

I. INTRODUCTION

The Internet of Things has completely changed how devices communicate and work together, but it has also brought about new security issues. With more and more IoT devices being used across different areas, making sure these devices are secure has become really important. Our paper discusses how we can use Convolutional Neural Networks to detect intrusions in IoT systems. The paper explores how CNN and its two configurations, VGG16 and Xception, work on an EdgeIIoTset that is a dataset of IoT and IIoT. By using CNNs, we were able to conclude that it can be a very flexible and effective way to enhance the intrusion detection capability of a system. Section 2 of the paper discusses the methodology used. Section 3 details the dataset utilized in the analysis. Following that, Section 4 delves into the attained results. Lastly, Section 5 provides an overview of the conclusions drawn.

A. CNN and its configurations(VGG19 & Xception)

Convolutional Neural Networks (CNNs), or generic CNNs, are a special kind of artificial neural network designed to work with structured grid data, like images. CNNs use several layers of convolution to pick out different levels of features

from images they're given as input. They gradually capture basic elements like edges and textures, then merge them to identify more complex patterns such as shapes and objects. VGG16 is an example of a CNN architecture that is simple and effective. Xception[4] is quite different from traditional CNN architectures like VGG16. It was developed by François Chollet. Xception adopts an approach called depthwise separable convolutions. Despite their differences, both VGG16 and Xception prove the effectiveness of CNNs in various image-related tasks. The paper presents how CNN, VGG16, and Xception work in the intrusion detection of EdgeIIoTset. The

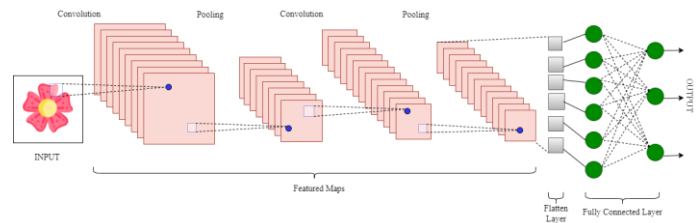


Fig. 1. Architecture of CNN[14].

comparison between CNN, VGG16, and Xception reveals distinct architectural features. While CNNs serve various image processing tasks with diverse architectures, VGG16 stands out with its 16-layer design, emphasizing stacked convolutional layers for feature extraction. In contrast, Xception employs depthwise separable convolutions to enhance efficiency by capturing spatial and channel-wise correlations. Pooling layers, although utilized in CNNs and Xception for dimension reduction, are a defining feature of VGG16, integrating max-pooling layers. Similarly, fully connected layers, common in CNNs, are prominent in both VGG16 and Xception, facilitating feature transformation. Activation functions, primarily ReLU-based, are uniformly applied across the architectures. Notably, Xception's innovative approach integrates depthwise separable convolutions and skip connections for improved efficiency and performance, distinguishing it from both CNN and VGG16 architectures. Figure 2 shows the configurations, VGG16 and Xception, of the CNN architectures:

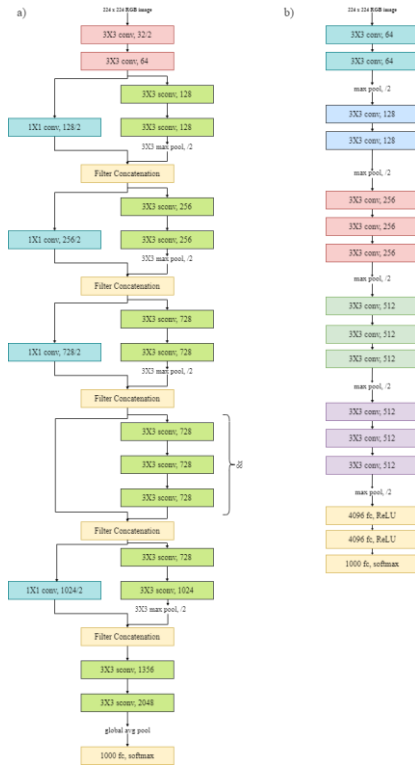


Fig. 2. a) Xception b) VGG16

II. METHODOLOGY

The methodology used in the paper involves data pre-processing, model training, testing and evaluation, and classification methods and results.

- Data transformation takes place in data pre-processing and training and testing images are generated.
- In model training, the model is trained.
- In testing and evaluation, data preparation is done, model loading takes place, predictions are made on the models, evaluation metrics are calculated, and a classification report is generated.
- Classification methods include CNN and its configurations (VGG16 & Xception). Figure 4 depicts the methodology used to conduct this study.

III. DATASET

Ferrag et al. [1] introduced the Edge-IIoTset, which includes ten different types of IoT devices. These devices cover a range of functions such as ultrasonic sensing, pH measurement, water level detection, soil moisture monitoring, heart rate tracking, flame detection, and digital sensing for temperature and humidity.

The dataset also has an analysis of 13 different attack classes, as shown in Figure 3. These attacks are categorized into five threats, including man-in-the-middle attacks, DoS/DDoS attacks, information gathering, malware attacks, and injection attacks. Furthermore, the dataset offers 61 features that have strong correlations. These features were

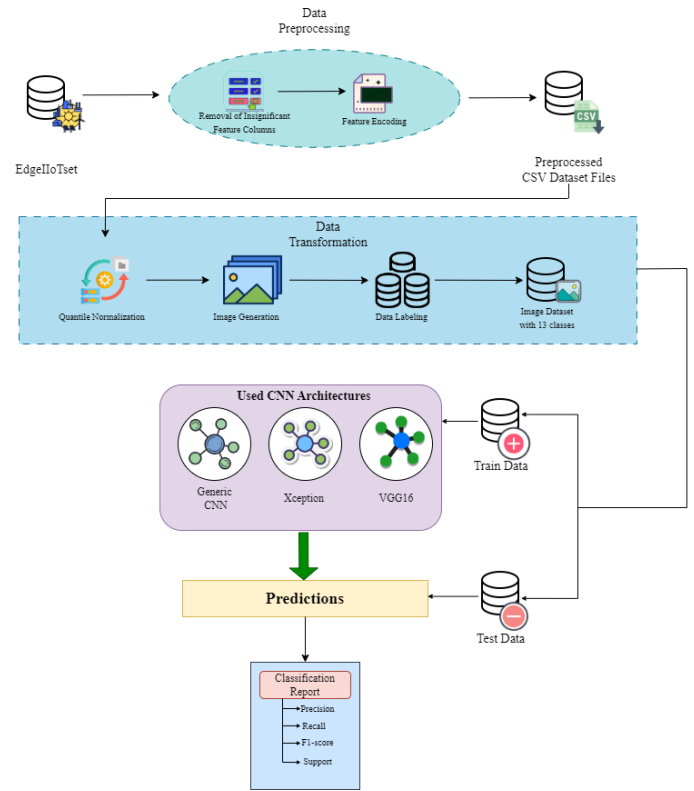


Fig. 3. Methodology Used

selected from a pool of 1176 found features, gathered from various sources like logs, alerts, system resources, and network traffic [16]. The testbed architecture proposed for generating the Edge-IIoTset dataset is structured into seven layers. These layers include the Cloud Computing Layer, Blockchain Network Layer, Fog Computing Layer, Edge Computing Layer, SDN Layer, Network Functions Virtualization Layer, and IoT and IIoT Perception Layer. The dataset was loaded, and then it was checked for any NaN values. There were no missing values in this dataset. Then, in the next stage, we got rid of 815 duplicate rows. Table I shows some of the unnecessary features that were dropped. Dummy encoding was applied to those feature columns that had object data types, resulting in 1909671 rows and 96 columns. Attack_Type was moved to the end. Then an image dataset was generated for each of the 13 attack classes. After that, the image dataset was split into a training set and a testing set. The images were adjusted in size to 224 by 224 pixels to improve the training of CNN models.

IV. RESULTS AND FINDINGS

The paper checked how well its suggested models worked by using certain measures of performance, as follows:

- Precision: measures the positive predictions made by the model.

$$P = \frac{TP}{TP + FP}$$

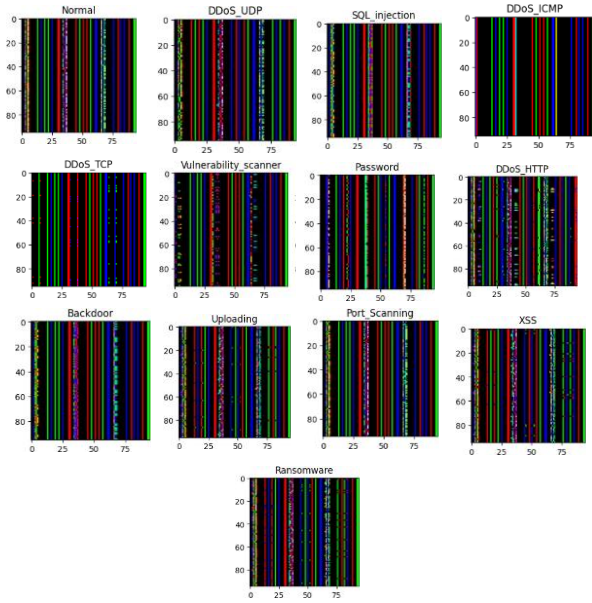


Fig. 4. Different classes of attacks

TABLE I. FEATURE DESCRIPTION

Features Used	Protocol Layer	Data Type
frame.time	Frame	Date and time
http.file.data	HTTP	Character string
http.request.full.uri	HTTP	Character string
http.request.uri.query	HTTP	Character string
ip.dst.host	IP	Character string
ip.src.host	IP	Character string
arp.src.proto.ipv4	ARP	IPv4 address
icmp.transmit.timestamp	ICMP	Unsigned integer
mqtt.msg	MQTT	Sequence of bytes
tcp.dstport	TCP	Label
tcp.payload	TCP	Sequence of bytes
tcp.srport	TCP	Unsigned integer
tcp.options	TCP	Sequence of bytes
udp.port	UDP	Unsigned integer
arp.dst.proto.ipv4	ARP	IPv4 address

- Recall: assesses the model's capability to accurately recognize positive cases.

$$R = \frac{TP}{TP + FN}$$

- F1-score: provides single score which balances both precision and recall.

$$F1 = \frac{2 \times P \times R}{P + R}$$

- Accuracy: is the ratio of instances that were classified correctly compared to the total number of cases.

$$TP + TN$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%$$

- Support: represents the number of instances in the dataset that belongs to each class.

At last, we calculate the weighted average of all performance metrics for overall performance. Generic Convolutional Neural

Networks and their configurations, namely VGG16 and Xception, were employed for classification, and then predictions were made. Tables II, III, and IV show how well the Generic CNN, Xception, and VGG16 models performed in identifying various types of attacks within the dataset. Generic CNN achieved the highest accuracy, topping the chart at 98.98%, followed by Xception with an overall accuracy of 96.49%, and VGG16 with an accuracy of 86.27%. Figures 5, 6, and 7 illustrate the accuracy and loss trends over the number of epochs for both the test and train sets of Generic CNN, Xception, and VGG16, respectively.

TABLE II. GENERIC CNN

	Precision	Recall	f1-score	Support
Backdoor	0.7917	1.0000	0.8837	19
DDoS HTTP	1.0000	1.0000	1.0000	38
DDoS ICMP	1.0000	1.0000	1.0000	43
Normal	1.0000	1.0000	1.0000	974
DDoS UDP	1.0000	1.0000	1.0000	83
Password	0.9643	1.0000	0.9818	27
Port Scanning	1.0000	1.0000	1.0000	17
Ransomware	0.0000	0.0000	0.0000	5
DDoS TCP	1.0000	1.0000	1.0000	36
SQL Injection	0.8857	0.9688	0.9254	32
Uploading	1.0000	1.0000	1.0000	16
Vulnerability scanner	1.0000	1.0000	1.0000	33
XSS	1.0000	0.6000	0.7500	10
Weighted Avg.	0.9898	0.9925	0.9906	1333
Macro Avg.	0.8955	0.8899	0.8878	1333

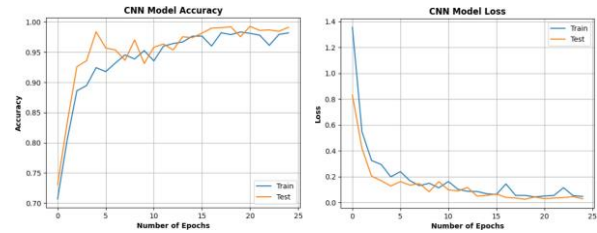


Fig. 5. Accuracy & Loss of Generic CNN

TABLE III. XCEPTION

	Precision	Recall	f1-score	Support
DDoS TCP	1.0000	1.0000	1.0000	36
Backdoor	0.7273	0.4211	0.5333	19
DDoS HTTP	1.0000	1.0000	1.0000	38
DDoS ICMP	1.0000	1.0000	1.0000	43
DDoS UDP	1.0000	1.0000	1.0000	83
Normal	1.0000	1.0000	1.0000	974
Uploading	0.9375	0.9375	0.9375	16
Password	0.7297	1.0000	0.8437	27
Port Scanning	0.6538	1.0000	0.7907	17
Ransomware	0.0000	0.0000	0.0000	5
SQL Injection	0.6111	0.6875	0.6471	32
Vulnerability scanner	1.0000	1.0000	1.0000	33
XSS	0.0000	0.0000	0.0000	10
Weighted Avg.	0.9649	0.9722	0.9670	1333
Macro Avg.	0.7430	0.7728	0.7502	1333

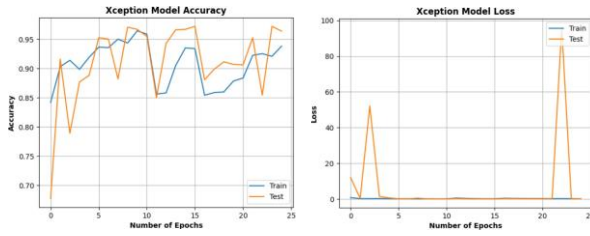


Fig. 6. Accuracy & Loss of Xception

TABLE IV. VGG16

	Precision	Recall	f1-score	Support
DDoS ICMP	1.0000	1.0000	1.0000	43
Backdoor	0.0000	0.0000	0.0000	19
DDoS HTTP	0.2289	1.0000	0.3725	38
DDoS TCP	0.0000	0.0000	0.0000	36
DDoS UDP	1.0000	1.0000	1.0000	83
Normal	1.0000	1.0000	1.0000	974
Password	0.0000	0.0000	0.0000	27
Port Scanning	0.0000	0.0000	0.0000	17
Ransomware	0.0000	0.0000	0.0000	5
SQL Injection	0.5000	0.3750	0.4286	32
Vulnerability scanner	0.7674	1.0000	0.8684	33
XSS	0.0000	0.0000	0.0000	10
Uploading	0.0000	0.0000	0.0000	16
Weighted Avg.	0.8627	0.8875	0.8676	1333
Macro Avg.	0.3459	0.4135	0.3592	1333

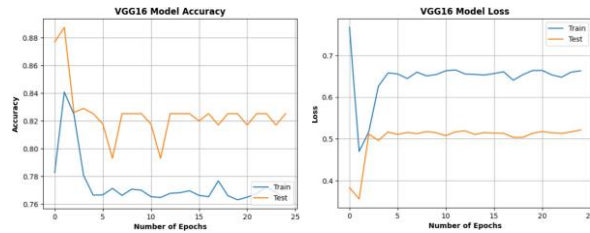


Fig. 7. Accuracy & Loss of VGG16

V. CONCLUSION

We might choose VGG16 because it's straightforward, easy to work with. On the other hand, Xception might catch your eye due to its efficiency, impressive performance, and knack for grasping complex features overall the findings suggest that using Convolutional Neural Networks for intrusion detection in edge IIoT environments could be an innovative approach, given the impressive results achieved: 98.98% for Generic CNN, 96.49% for Xception, and 86.27% for VGG16, respectively. Each CNN model was trained for approximately 25 epochs, with accuracy peaking as epochs increased and eventually stabilizing. Despite these promising outcomes, deep learning-based intrusion detection in IoT still faces notable challenges like data imbalances, limited computing power, and the diversity of IoT systems. Exploring ways of improving DL-based intrusion detection in IoT settings is an area of interest for researchers.

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